

Sharma_DSC520Week4.2

2024-12-21

```
# 1. Student Scores and Counts are the observational units in this dataset.
# 2. Based on Paragraph Naratives 'Section' (Categorical) and 'Course Grades'
# and 'Total Point earned' (quantitative)

# Read CSV file for student scores in different section
student_scores = read.csv("scores.csv")

library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

regular_section <- filter(student_scores, Section == "Regular")

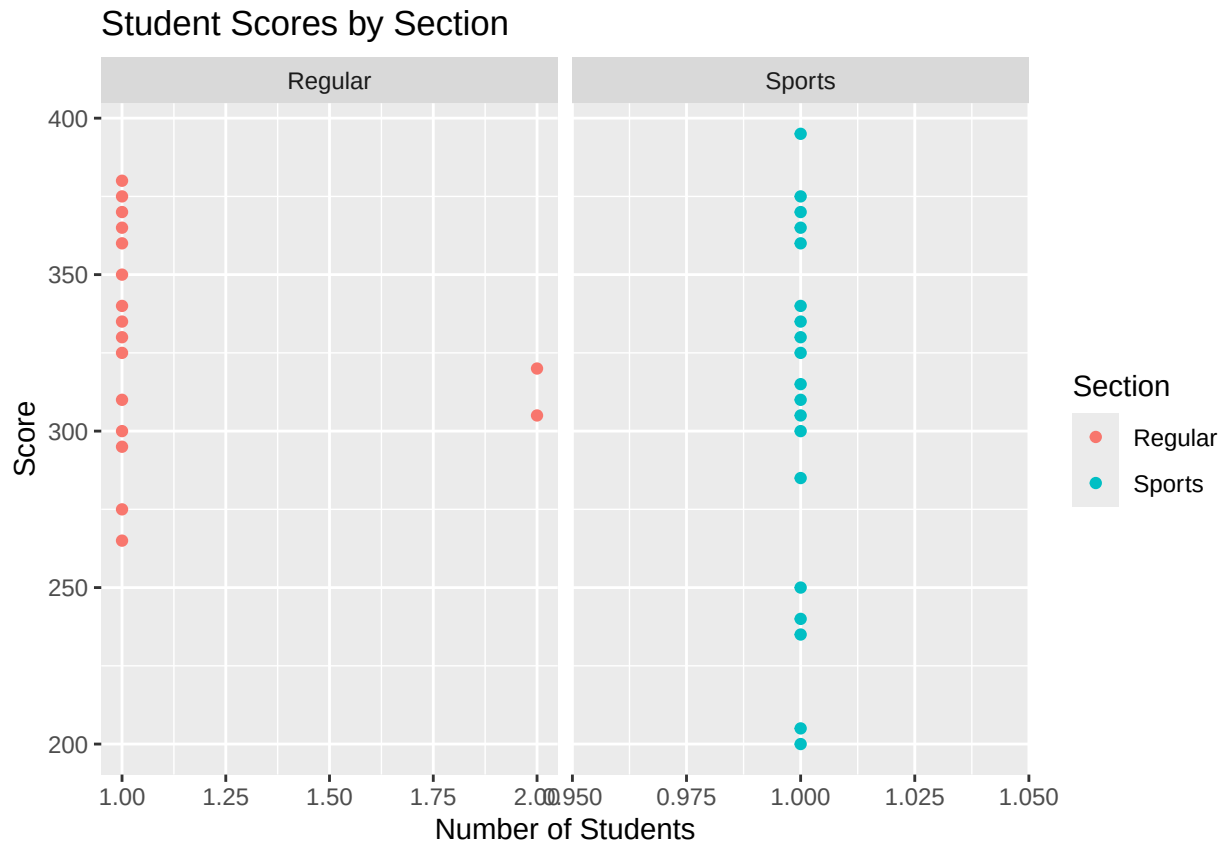
sports_section <- filter(student_scores, Section == "Sports")

library(ggplot2)
library(dplyr)

# Create a table of counts for each score in each section
section_scores <- student_scores %>%
  group_by(Section, Score) %>%
  summarise(Count = n())

## `summarise()` has grouped output by 'Section'. You can override using the
## `.groups` argument.

# Plot the data using the plot function
ggplot(section_scores, aes(x = Count, y = Score, color = Section)) +
  geom_point() +
  facet_wrap(~ Section, scales = "free_x") +
  labs(title = "Student Scores by Section", x = "Number of Students", y = "Score")
```



*# Q1. Comparing and contrasting the point distributions between the two section,
looking at both tendency and consistency: Can you say that one section tended
to score more points than the other? Justify and explain your answer.*

*# Ans1. Yes and Sports section tending towards more points where as more
students tending towards higher points.*

*# Q2 Did every student in one section score more points than every student in
the other section? If not, explain what a statistical tendency means in this
context.*

*# Ans2. statistically students in both the sections are varying differently and
in sports section students are getting ≤ 250 and also ≥ 250 where as in
regular section students are only getting scores more than 250 or so.*

*# Q3 What could be one additional variable that was not mentioned in the
narrative that could be influencing the point distributions between the two
sections?*

*# Ans3. there could be multiple variables related to students i.e. Age, gender
and other demographic variables.*